Food Access in Bay Area Neighborhoods

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Research Question

Does food access and public health burden differ across Bay Area neighborhoods? How?

Our scope now focuses on a narrower geography with the following selected Bay Area counties: Alameda, Contra Costa, and SF. To keep time frames as consistent and recent as possible, we will use the most recent data available: 5 yr ACS for Census Data (2020-2024), CDC PLACES 2025 release.

Our analysis covers several domains of 'food access,' including travel times to stores, store types, and cultural and dietary preferences.

Data & Cleaning

Public Health Data

We used the CDC PLACES 2025 (which uses health estimates from 2023) census tract release data for adult obesity, diabetes, coronary heart disease, stroke, and high blood pressure. Since the full dataset was very large, we used the CDC API to retrieve the data for our study area. We kept GEOIDs as strings so their leading zeroes would not be lost, and then matched the health data to the selected census tracts using GEOID. We kept missing estimates as missing values rather than filling them in. The current file contains 1668 tracts and 8 tracts are missing at least one health estimate. We kept those tracts in the dataset and left the unavailable estimates blank so as to not lose any particular tract's information.

Census Data

For census data, we cleaned an initial dataset of 865 census-tract-level records from the US Census Bureau, filtering for Alameda, Contra Costa, and SF counties. To get this dataset, we used an API call to the ACS 5-year (2020-2024) data and merged it with MTC's clipped SF Bay Area Census tract shapefiles to get geometry for mapping.

First, we standardized column names and data types. ACS variable codes were renamed to their respective descriptive labels, numeric variables as integers or floats, and geographic identifiers as strings. We removed six census tracts with a population of zero (approximately 0.7% of the original dataset), as they are not applicable to our analysis of populated areas and the redundant column 'countyfp' from the join. This leaves us with 859 tracts.

To create a visualization-friendly dataframe ('viz'), we replaced 35 invalid Census data points, coded as sentinel values for aggregated estimates (10 household-income, 21 per-capita income values, 2 aggregate-vehicle, 2 avg-household size) to NaN. These are unavailable estimates, so we left them out because they can distort summary statistics and visualizations. The aggregate vehicle column had 24 initial missing values (to 26 after cleaning for sentinel codes), but we estimated and filled in 24 of these using the household vehicle columns from the initial census call. We assigned 4 to households with 4+ vehicles; thus, the estimate may undercount actual totals and was flagged in a separate column. We also kept and flagged household incomes top-coded at $250,001. We also mapped county codes to readable county categories.

Identifying Store Places

Originally, we intended to use OSM for all store places data. However, querying was slow and results had significant missing information. Therefore, we turned to a more robust commercial alternative: Google Maps. We used Google Maps Places API to perform a one-time harvest of all grocery stores, convenience stores, and liquor stores in San Francisco and the East Bay. Using Kikuchi et al (2026) as inspiration, we used an adaptive grid search approach to minimize the number of calls to Google Maps API. Using this approach, we first overlay large squares over the Bay and find how many tract centroids are bounded within them. Squares containing greater than 6 centroids are recursively split into 'children' until all squares contain less than 7 tract centroids. Cells containing no land (IE are floating in the bay) are discarded, and remaining cells are assigned a scaled search radius between 100 meters and 50 km to cover the entirety of their land mass. This approach naturally minimizes API calls by producing smaller cells in dense urban areas and larger ones in sparse regions, saving time and credits.

After assigning coordinates for sample points at the center of each search circle, we ran a Google Maps Places 'nearby' search for the target types: grocery_store, convenience_store, and liquor_store. The API limits results to 20 per search, so if one call produced 20 results, circles were further split into children and queried again so that as few stores were missed as possible. Taking advantage of Google Maps functionality and wealth of information, we also included the following attributes for each found place: place_id, name, lat, lon, rating, usering_ratings_total, vicinity, address, price_level, business_status, primary_search_type (our search type), primary_type (Google's main definition for a place), types (additional subtypes to describe a location), and found_by (includes all search terms that recovered this place). Results were stored in bay_food_places.csv inside the data directory. It is worth noting that the file that generated this data, harvest_food_places.py, does not need to be run again for our final notebook; if it was run again with the same API key, it might cost real dollars!

Next, we performed several cleaning steps on bay_food_places. Our harvest scanned through 862 tracts and returned 3350 stores. Many results included extraneous 'primary_types', including restaurants, beauty salons, hardware stores, and etcetera. 3066 stores remained after filtering out extraneous store types. Additional EDA revealed outliers including smokeshops, health and vitamin stores, and incorrectly labeled primary_types, for example. After fine-toothed filtering and deduplication of identical addresses and store names, 2911 stores remained, divided into 669 liquor stores, 890 convenience stores, and 1352 grocery stores. Within each primary_type subcategory, there are further divisions into gas_stations, delis, asian_grocers, and etcetera. It is likely that some stores may not perfectly sit in one primary_type label (ex: small community markets might not be as well stocked as a commercial grocer, but have more produce than a convenience store), but spot analysis confirms that Google Maps primary_type labeling is generally an adequate descriptor of a store's primary function for these preliminary analysis.

Analytical Methods

Health Indicator Preparation and Mapping

We joined the health estimates to the census tract boundaries by matching GEOIDs. This let us map each estimate by tract. We made a map for each health indicator and used county boxplots and histograms to compare how the estimates vary. The correlation heatmap shows how the health indicators vary together, although it does not yet compare health with food access.

Distance to Food Location

To represent each tract at least once, we created a 500 meter-spaced grid covering the Bay (excluding water). We then spatially joined this grid to tract geography information. At each point, we used a SciPy cKDTree search to find every store within walking (0.5 miles) and driving (5 mile) distance. We stored these results, grouped them by geoid, and found the average number of stores within either walking or driving range. Results for each store type (liquor, convenience, and grocery) by tract are plotted below.

Cultural Food Location Filtering

If we wanted to know cultural food accessibility, we needed to first figure out which areas were in need of cultural food staples. To do this, we looked at the tracts that met criteria for certain ethnicities. For now, we selected Hispanic, South Asian, East Asian, and Muslim. We looked at tracts that had at least 250 people of that ethnicity in the population. We also wanted to account for areas that had significantly large populations, so we allowed for tracts that had greater than or equal to 5% of the population. The Census did not have information on religion, so we needed to figure an alternative method for figuring that out. So we used ancestry from countries that have majority Muslim populations. This resulted in no tracts meeting the requirement, so we used all tracts for them.

We then needed to label each food location with the cultural foods they provided. The food location dataset had a types column which gave us a general idea of what the food location was for. From here, we selected the relevant types that help classify them like “asian grocery store.” But we found this to be somewhat lacking. So we parsed through each name of the food locations and created a list of all terms. We removed numbers, single-digit items, etc. We made certain exceptions like 7-eleven. There were way too many terms to sort through by hand, so we sampled through the food locations by 20s, adding the terms to their respective ethnic lists. We repeated this numerous times.

We then use our distance to the grocery process for each ethnicity we are studying, with their respective tracts and food locations. We want to eventually combine this with our distance to all-grocery store data and also include other metrics that are diet-specific and more.

Mapping Data

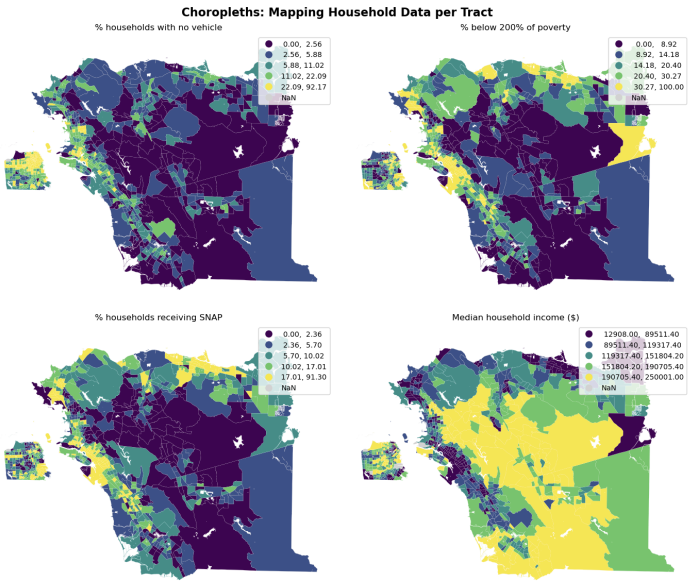
For census data, geography coordinates were reprojected from EPSG: 4326 to EPSG:3310 for California-specific spatial analysis.

For map distances, tract geographies and store locations were reprojected from EPSG: 4326 to EPSG: 26910 to work in meters.

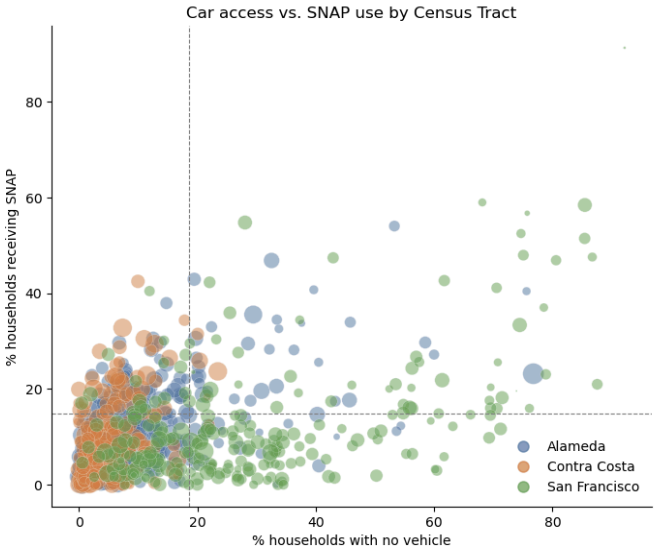
Early Results

Census Analysis

These choropleths show geographic clusters where many tracts with higher poverty & SNAP participation also have a lower household income. Additionally, limited vehicle access overlaps with some of these areas but does not show the same pattern, suggesting that vehicle



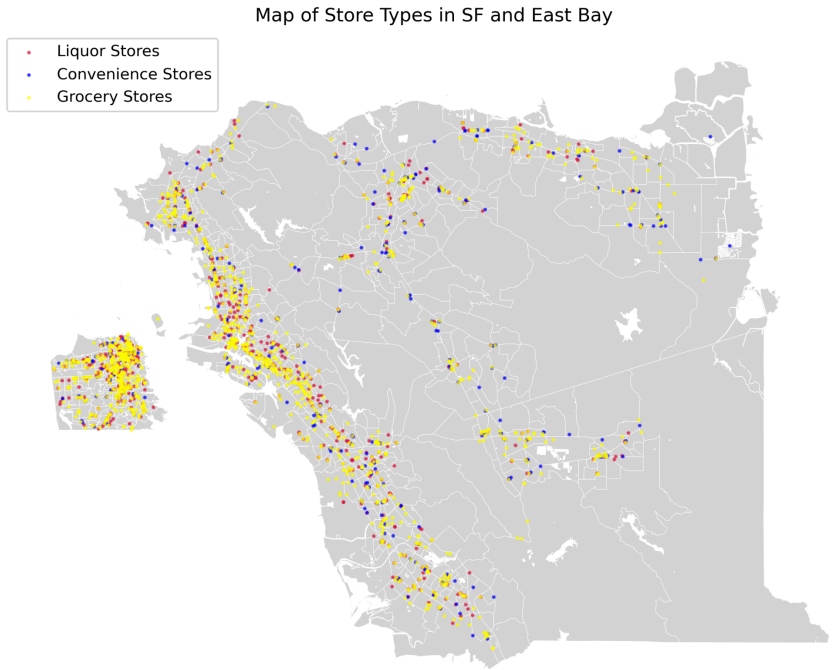
access and economic disadvantage may be related but not consistently. This process may help inform which specific tracts to further look into and where our analysis might be most impactful.



Demographic-level charts might not be as applicable right now for analysis until we target relevant tracts and then look for any racial disparities, age mobility-related issues, etc.

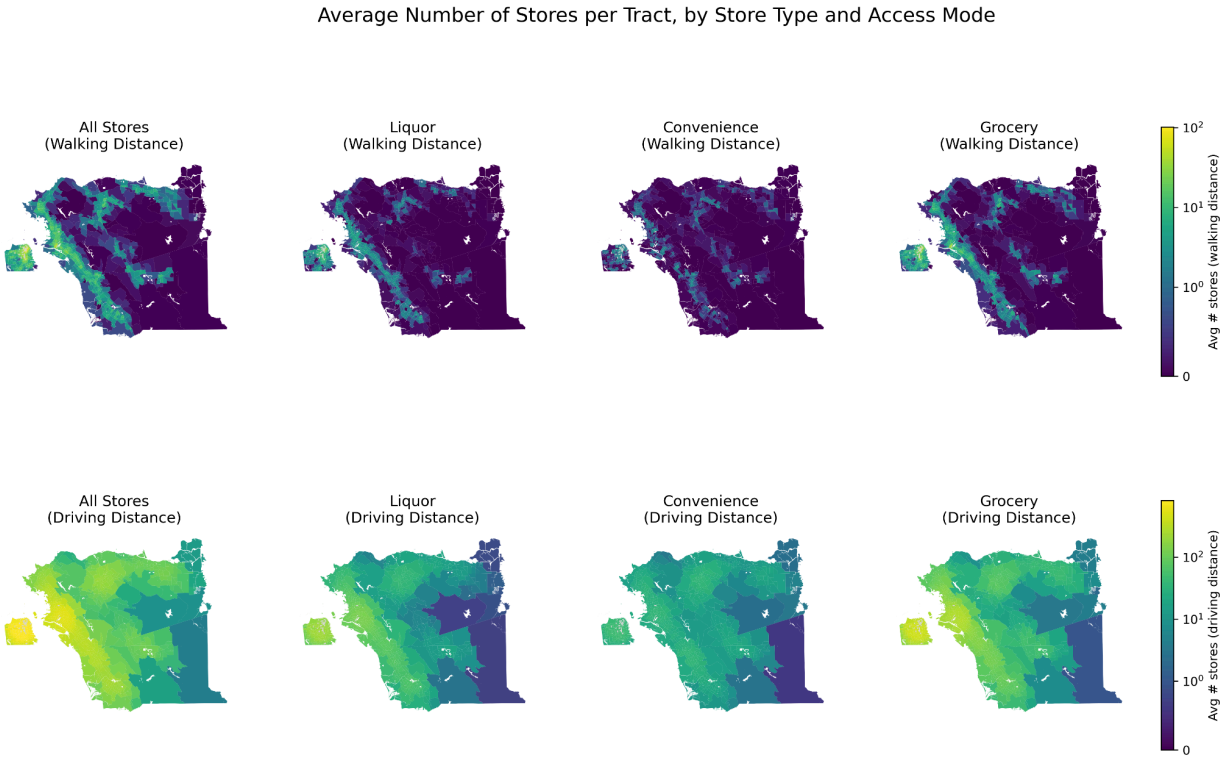
The top-right quadrant of the scatter plot to the left, highlights areas with a high percentage of SNAP households with no vehicle, specifically in Alameda and SF. It might help to make this chart interactive for future exploration, since it could help identify tract-level characteristics.

It is surprising that limited vehicle access is not quite consistent with higher SNAP, though we may be able to explain this by exploring the data further with other modes of transportation, like public transit and distance data.



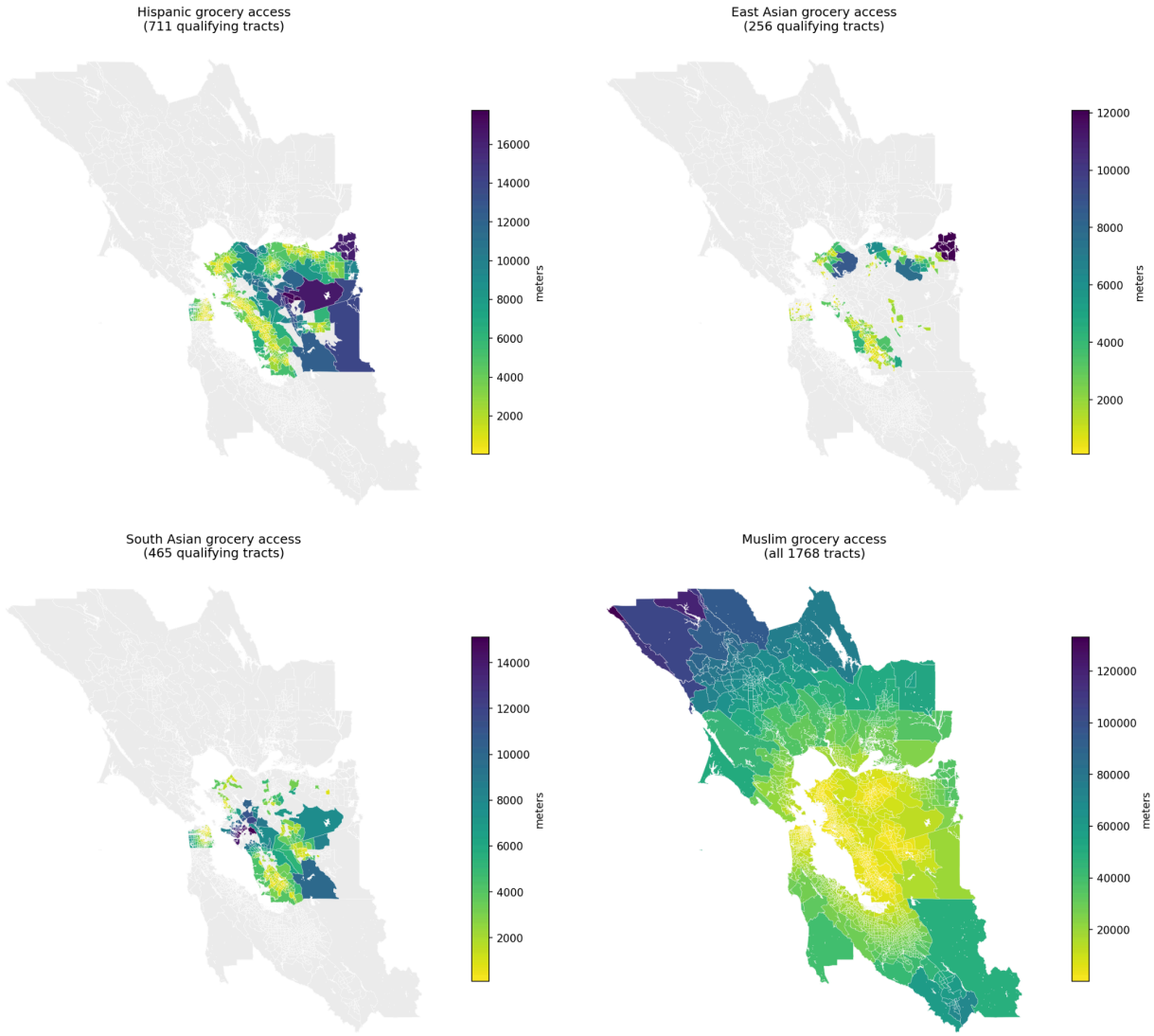
Food Walking & Driving Distance Analysis

The map to the left shows all stores included after harvesting and filtering. One can see the many regional preserves and wilderness areas along the mountains. More populous areas like SF and Oakland have an evident higher density of stores, whereas more suburban areas like Livermore have a marked lower density of stores.



These chloropleths show which tracts have more store types in walking and driving distance to them relative to others. This first pass uses a ‘buffer’ approach, which is less computationally expensive at the cost of travel time accuracy. Eventually, for each point, we will define ‘walking distance’ as a walk that takes 15 minutes or less according to Google Maps, ‘biking distance’ as a trip taking 30 minutes or less by bicycle, and ‘transit distance’ as a trip taking 30 minutes or less by transit according to Google Maps. These preliminary results do in fact show major differences between tracts. San Francisco, by nature of having a smaller area, appears the brightest, but does have some gaps in walking distance access. In the East Bay, liquor stores can be seen cropping up more in urban areas relative to convenience stores. While current results might mimic population density metrics, further analysis using travel time instead of travel radius will hopefully reveal more meaningful patterns in store accessibility and quality. Additionally, comparing ratios of convenience, liquor, and grocery stores to one another will reveal which tracts are food ‘swamps’ (IE less full-service grocery stores, more prepared-food and convenience style stores).

Cultural Food Accessibility Analysis



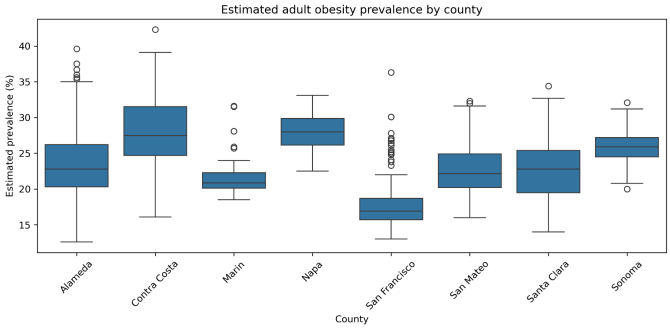
We can see that the places with the most access are San Francisco County and the West of Alameda County. This makes sense given our own domain knowledge. SF, Berkeley, and Oakland are full of these stores. But as we get further away from the area, we see that cultural food accessibility weakens. To note, there is an area to the eastmost of where we are studying that are particularly inaccessible. This area is full of islands, which may explain that. It's important to note that they do follow the general pattern of being more accessible as you are closer to the coast, but there are some differences.

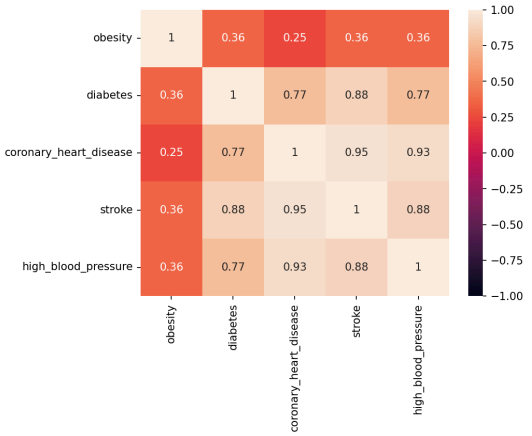
It is important to note that there are not many tracts that qualify in the east of Contra-Costa and Alameda County. We may want to consider lowering the requirements for consideration.

Health Indicator Analysis

The maps and charts show differences in estimated health prevalence across tracts. The highest obesity estimate in the current table is 42.3% in Contra Costa County. One San Francisco tract has especially high estimates for diabetes, coronary heart disease, stroke, and high blood pressure. We will check that tract

and the CDC uncertainty estimates before interpreting it as a neighborhood pattern. We have not yet combined the health estimates with food data, so we cannot tell from these results whether food access and health prevalence are related.





Plan to Finish

Timeline	Milestones	Group Members
Week 12: 11/9-13	Add Diet metric Analyze census data further to finalize specific tracts for focus Finalize health data	Kwang Robin Hafsa Vianna
Week 13: 11/16-20	Food Mirage Metric Match health data with food-access metrics Recalculate walking, transit, biking and driving distances with OSM based on *time* instead of birds-eye distance; compare against car ownership by tract	Kwang Vianna Robin Hafsa
Week 14: 11/23-27	Combine the diet, cultural, food mirage, distance metrics to create a new definition for food deserts. Compare food access, health, census & identify patterns Calculate food ‘desert’ and food ‘swamp’ metrics	Kwang Vianna, Hafsa Robin
Week 15: 11/30-12/4	Attempt PCA to reduce features over all tracts into 2 dimensions Github Pages Website	Robin Kwang Hafsa Vianna

Week 16: 12/7-11 (RRR)	Catch up on anything that hasn't been finished yet. Organize GitHub repository with relevant files Attempt to develop an app that recommends food options.	Everybody
Week 17: 12/13-15	Finishing Touches Tue, Dec 15: Submit Project!	Everybody

Risks to finishing/desired help:

1. When we are working on the cultural, diet, and food mirage metrics, we need information on what the food location provides. There is no data for these food locations on what produce they sell. To figure this out, we need to parse through the names of the food locations and get all the possible types. Then we need to go through all of them to label them. So far, we have only sampled the possible labels and then did the labeling process with them. So we need to develop a better process to do this. Worst case scenario, we brute force and label all of them by hand.
2. We have plans to attempt 1) a PCA over all tracts simplifying details about store type, accessibility, culture, affordability, and health to see if the resulting principal components act similarly to 'food desert' indices or possibly some other pattern. However, the data is all very different, and while we can normalize it, we may struggle to interpret results. Worst case we won't include the analysis; we can also try clustering instead if PCA doesn't work.
3. We were excited to use the affordability metric from Google Maps as a feature of our analysis but many entries turned out to be missing. We don't have a good way to infer affordability outside of this and scraping the internet. To address this, we may have to infer based on online reviews of larger stores, but this might leave out smaller stores that do not have as many reviews.

- Is there a better way to figure out classifying each store for whatever metric we are examining?
    - Ex: For the distance to stores analysis, we are currently dividing the Google Maps generated data into 3 discrete groups – grocery, liquor, and convenience. We would like to know if there is an automated/semi automated way to further define stores into 'box' or 'chain' stores versus smaller, local brands. Generally we are seeking out the niche that each store plays in its neighborhood. Tips towards quantifying this are welcome.